VAYUCOUPLER

Air Pollution–Weather Coupled Forecasting & Predictive GRAP Decision Support System

PROBLEM STATEMENT

- **ID:** SIH26082 (Software Edition)
- **Title:** Air Pollution–Weather Coupled Forecasting (Delhi NCR Focus)
- **Ministry:** Ministry of Earth Sciences (MoES)
- **Alignment:** CAQM, CPCB, IMD & NCR States

CORE VALUE PROPOSITION

- **Lead Time:** 48h to 72h Pre-emptive Action Warning
- **Physics:** Explicit Boundary Layer & Inversion Coupling
- **Telemetry:** NASA FIRMS Stubble Vector Corridor (315°)
- **Simulation:** Real-time Counterfactual 'What-If' Engine

TEAM ATOMX & DEPLOYMENTS

- **Team Name:** AtomX (Lead: Vivek Raj)
- **Live Web:** vayucoupler.vercel.app
- **GitHub:** vivek-glitch15/VayuCoupler
- **Deployments:** Web Vercel + Windows Offline + Android APK

TEAM ATOMX DOMAIN SPECIALIZATION & ROLES:

- **Vivek Raj (Team Lead):** Full-Stack Architecture, Atmospheric Physics & Geospatial Engine
- **Member 2:** Planetary Boundary Layer Height (PBLH) & Thermal Inversion Modeling | • **Member 3:** Backend Telemetry & Multi-Agency Dispatch Architecture
- **Member 4:** Geospatial UI/UX, Particle Wind Flow & Visual Command Dashboards | • **Member 5:** ML Forecaster & Real-Time Source Apportionment Engine
- **Member 6:** Cross-Platform Mobile Android APK & Single-File Windows Desktop Offline Packaging

The Problem: Delhi's Smog Trap & The Fatal Flaw of Reactive GRAP

Every winter, 30 million citizens choke because existing emergency systems enforce curbs only after the crisis arrives.

■ FATAL FLAW OF REACTIVE GRAP

• **Post-Facto Trigger Rule:** CAQM Stage III/IV emergency curbs (halting construction, banning interstate trucks) are triggered ONLY AFTER ground stations record 'Severe' (400+) for 48 consecutive hours.
• **Zero Lead Time:** By the time diesel trucks or construction are halted, the dense smog trap has already formed over Delhi.
• **Irreversible Health Harm:** Children, asthmatics, and seniors inhale toxic particulate matter for 2 to 3 days before authorities react.
• **Economic Shock:** Sudden overnight blanket bans leave daily-wage workers stranded without preventing the initial spike.

■ DISCONNECTED METEOROLOGY

• **Trapping Crisis:** Delhi's air crisis is fundamentally an atmospheric physics problem, not merely an emission issue.
• **Boundary Layer Collapse:** Planetary Boundary Layer Height (PBLH) collapses from 1,800m in daytime to under 350m at night.
• **Thermal Inversion Lid:** Colder surface air trapped under warm aloft air prevents vertical dispersion ($\Delta T > 4°C$).
• **Legacy Black-Box Flaw:** Existing models treat AQI as an isolated statistical curve, ignoring real-time atmospheric dynamics.

■■ TRANS-BOUNDARY SILOS

• **Trans-Boundary Influx:** 30% to 45% of peak winter $PM_{2.5}$ in Delhi arrives via trans-boundary transport from Punjab & Haryana stubble burning.
• **Inter-State Blame Game:** Delhi declares red alert while upwind states continue burning due to lack of synchronized early warnings.
• **No 'What-If' Capability:** Current portals display static graphs; policymakers cannot simulate policy impacts before enforcing curbs.
• **No Role-Based Dispatch:** Agencies receive generic PDFs instead of automated, legally binding executive work orders.

Our Solution: Predictive GRAP Grounded in Atmospheric Physics

Zero black-box obscurity — coupling boundary layer compression with satellite fire telemetry for 48h–72h lead time.

THE PARADIGM SHIFT: REACTIVE GRAP ➜ PREDICTIVE GRAP (48h–72h ADVANCE LEAD TIME)
Calculates boundary layer collapse and stubble plume influx to trigger Stage II, III & IV curbs 2 to 3 days before smog traps settle.

1. VENTILATION INDEX (VI) — FLUSHING CAPACITY

`VI = Wind_Speed (m/s) × PBL_Height (m) [m²/s]`

- **VI > 3,500 m²/s:** Favorable dispersion; ground pollutants flush out rapidly.
- **VI < 2,000 m²/s:** Critical Trapping; Delhi basin behaves as an enclosed container.
- Nighttime boundary layer collapse causes 3× to 5× pollutant density accumulation.

2. THERMAL INVERSION TRAPPING (K_trap)

`K_trap = 1.0 + 0.38(ΔT_inv) + 1.4 × max(0, (2500 - VI)/2500)`

- Quantifies the thermal 'atmospheric lid' formed over Delhi NCR on cold winter nights.
- **ΔT_inv:** Temperature differential between 1,000m layer and surface.
- When K_trap > 2.2, standard emissions produce 'Severe+' emergency spikes.

3. UPWIND STUBBLE TRANSPORT VECTOR (S_vector)

`S_vec = FireCount × max(0, cos(θ_wind - 315°)) × (WS / 5.0)`

- Directional dot product of wind angle with the NW stubble plume corridor (315°).
- Only fires with aligned north-westerly wind vectors are transported into Delhi.
- Directly computes trans-boundary mass influx (µg/m³) rather than static tallies.

4. COUPLED FORECASTER & SOURCE ATTRIBUTION

`AQI(t+Δt) = F_phys(AQI_t, VI, K_trap, S_vec) ± 90% CI`

- Dynamic Source Apportionment: Stubble (38%), Vehicular (32%), Industry (15%), Dust (15%).
- Outputs calibrated +24h, +48h, and +72h continuous predictions with uncertainty envelopes.
- Directly triggers CAQM rules.json to generate automated, role-specific action tickets.

End-to-End 4-Tier Pipeline: Ingestion to Multi-Platform Delivery

Engineered for 100% offline hackathon demonstration resilience and frictionless cloud deployment.

LAYER 1: DATA INGESTION & DUAL-MODE ADAPTER

- **16 CPCB Ground Stations:** Anand Vihar, IGI, Punjabi Bagh, RK Puram, Jahangirpuri, Wazirpur, Okhla, etc.
- **IMD High-Altitude Meteorology:** Radiosonde boundary layer heights & Eulerian WRF wind fields.
- **NASA FIRMS Satellite:** MODIS/VIIRS thermal anomalies & Punjab/Haryana stubble counts.
- **Dual-Mode Adapter:** Live REST API integration + 168-Hour Synthetic Episode generator for 100% demo uptime.

LAYER 2: COUPLED ATMOSPHERIC PHYSICS & ANALYTICS ENGINE

- Physics Ventilation Modulator calculates real-time Ventilation Index (VI) & Inversion Trapping Factor (K_trap).
- Stubble Plume Directional Vector Projection (NW 315° corridor) • Dynamic Source Apportionment Engine.
- Gradient Boosting + Ridge regression forecaster generating continuous +24h, +48h, and +72h predictive envelopes with 90% confidence bounds.

LAYER 3: PREDICTIVE GRAP & DISASTER DISPATCH ENGINE

- Automated Predictive GRAP Rules Engine (evaluates forecast curves against configurable rules.json matrix).
- Multi-Agency Action Dispatcher generates role-specific payloads (Police, Agri, MCD, Schools, Hospitals).
- 'What-If' Counterfactual Policy Simulator for real-time Supreme Court & CAQM scenario testing.

LAYER 4: CROSS-PLATFORM DELIVERY SUITE & COMMAND CENTERS

- **MoES Command Center Web App:** Production on Vercel: https://vayucoupler.vercel.app
- **Windows Desktop Offline Edition:** Single-file standalone HTML executive command center (zero install).
- **Native Android Standalone APK:** 5.4MB field inspection app with offline caching and push dispatch.

Comparison Matrix & Interactive 'What-If' Policy Simulator

Proving technological superiority over SAFAR and empowering CAQM with empirical scenario testing.

CAPABILITY	LEGACY / SAFAR	VAYUCOUPLER (ATOMX)
GRAP Execution	Reactive (After 48h spike)	PREDICTIVE (48h–72h Lead Time)
Physics Coupling	Basic statistical / None	EXPLICIT (PBLH + Inversion + Vector)
Disaster Dispatch	Passive public PDF upload	AUTOMATED (6-Agency work orders)
Policy Simulation	Not Supported	INTERACTIVE 'WHAT-IF' SIMULATOR
Offline Resilience	Cloud-only dependency	100% OFFLINE (Windows & Mobile APK)

CLOSED-LOOP AUTOMATED DISPATCH MATRIX:

- **Punjab Agri:** Deploy 1,200+ Happy Seeders 48h before wind shift.
- **Traffic Police:** Divert 15,000+ freight trucks to EPE/WPE bypass.
- **MCD:** Anti-smog misting guns to Anand Vihar, Mundka & Wazirpur.
- **Health/Schools:** 24h advance notice for online classes & ICU readiness.

■ 'WHAT-IF' COUNTERFACTUAL SIMULATOR

Authorities dynamically slide policy interventions to preview resulting AQI curves before enforcing disruptive emergency bans.

SCENARIO A: STATUS QUO (REACTIVE GRAP)

- Monitors reach 400 at T-0h; curbs enforced only at T+48h.
- **Result:** Peak AQI reaches **485 (Severe+ Emergency)**. Schools shut abruptly, hospitals overrun with respiratory distress.

SCENARIO B: VAYUCOUPLER INTERVENTION

- Model predicts 485 spike 48h prior due to boundary layer collapse to 320m.
- Pre-emptive action: 45% truck bypass + 40% stubble reduction.
- **Result:** Peak AQI capped at **382 (Managed)**. Severe+ emergency completely averted!

Measurable Impact, National Roadmap & 3-Minute Demo Workflow

Delivering India's first operational Weather–Pollution Coupled Predictive GRAP System.

48 to 72 HOURS
ADVANCE ACTION LEAD TIME
Transforms panic emergency bans into structured 3-day readiness.

25%–35% REDUCTION
FEWER HOSPITALIZATIONS
Shields vulnerable children, seniors & asthmatics from severe spikes.

■1,200+ CRORE
AVOIDED ECONOMIC LOSS
Prevents chaotic overnight industrial & freight shutdowns.

■■ SCALABILITY & MOES ADOPTION ROADMAP

- **Phase 1 (Months 1–3) | Delhi NCR Pilot:** CAQM integration with 40 CPCB CAAQMS stations + daily automated predictive briefings dispatched to Commission members.
- **Phase 2 (Months 4–8) | Indo-Gangetic Basin:** Scaling to Kanpur, Lucknow, Patna + INSAT-3D AOD & Sentinel-5P TROPOMI trace gases.
- **Phase 3 (Months 9–12) | NCAP Integration:** Direct coupling with IMD's 3km Eulerian WRF-Chem and CEMS across 1,000+ industrial smokestacks. Zero license lock-in.

■■ 3-MINUTE JUDGE DEMONSTRATION WORKFLOW

- **0:00 - 0:45 | Elevator Pitch:** Open vayucoupler.vercel.app. Pitch: 'Current GRAP is reactive; VayuCoupler predicts spikes 48h early via meteorology coupling.'
- **0:45 - 1:30 | 7-Day Scrubber:** Scrub to T-72h. Show monitored AQI is still Moderate (~210), but boundary layer collapses to 340m, forecasting 480+ for T+48h.
- **1:30 - 2:15 | Predictive GRAP:** Show Stage III/IV in 'PRE-EMPTIVE TRIGGER' state. Inspect automated dispatch payloads for Punjab Agri & Delhi Police.
- **2:15 - 3:00 | 'What-If' Testing:** Move Stubble Reduction to 50% and Truck Diversion to 40%. Watch peak AQI drop from 485 to 382. Ready for Jury Q&A;!